## Classical XY model

$$H_{XY} = -J \sum_{\langle nm \rangle} \cos(\varphi_n - \varphi_m) = -J \sum_{\langle nm \rangle} \vec{s}_n \cdot \vec{s}_m = -\frac{J}{2} \sum_{ng} \vec{s}_n \cdot \vec{s}_{n+g}$$

where classical unit vectors $\vec{s}_n = (\cos\varphi, \sin\varphi, 0)$ are planar (!)

(This can be obtained as a limit of $\vec{N}$ by restricting $\theta = \frac{\pi}{2}$ : $\vec{N}_n = (\cos\varphi \sin\theta, \sin\varphi \sin\theta, \cos\theta) \xrightarrow[\theta=\frac{\pi}{2}]{} \vec{s}_n = (\cos\varphi, \sin\varphi, 0)$.)

Partition function

$$Z_{XY} = \int \prod_r d\varphi_r \, \exp - \frac{H}{k_B T}$$

① High temperature $T \gg J$ : disordered phase

② Low temperature : ordered (as we will see later)

In the ordered phase spins at neighboring sites are well correlated $\Rightarrow$ can approximate cosine by its expansion : (continuum approximation)

$$H_{XY} = -\frac{J}{2} \sum_{ng} \cos(\varphi_n - \varphi_{n+g}) \approx -\frac{J}{2} \sum_{ng} \cos(\vec{g} \cdot \vec{\nabla}\varphi) =$$

$$\varphi_{n+g} \approx \varphi_n + g_x \frac{\partial \varphi_n}{\partial x} + g_y \frac{\partial \varphi_n}{\partial y} = \varphi_n + \vec{g} \cdot \vec{\nabla}\varphi$$

$$\cong -\frac{J}{2} \sum_{ng} \left\{ 1 - \frac{1}{2} \left( g_x \partial_x \varphi_n + g_y \partial_y \varphi_n \right)^2 \right\} = \text{const}$$

$$+ \frac{J}{4} \sum_n \sum_g \left\{ g_x^2 (\partial_x \varphi_n)^2 + g_y^2 (\partial_y \varphi_n)^2 + 2 g_x g_y \, \partial_x \varphi_n \, \partial_y \varphi_n \right\}$$

$$\vec{g} = \begin{cases} a\,(\pm 1, 0) \\ a\,(0, \pm 1) \end{cases} \Rightarrow \sum_g \text{ removes last term}$$

$$H_{xy} \approx J a^2 \sum_h (\partial_x \varphi)^2 + (\partial_y \varphi)^2 = J \int dx dy \, (\vec{\nabla} \varphi(r))^2$$

$$\vec{r} = (x, y)$$

(Compare again with $S_N(\theta = \frac{\pi}{2}) = \frac{1}{2g} \int dx dy \, (\vec{\nabla} \varphi)^2$ !)

$$Z_{xy} = \int \prod_r d\varphi(r) \, \exp - \frac{J}{k_B T} \int dx dy \, (\vec{\nabla} \varphi(r))^2 .$$

Mathematically this is identical to the quantum model (with $g \to \frac{k_B T}{2J}$)

but only if one neglects topological phases. (which of course are absent classically)

By understanding $Z_N$ we will learn about $Z_{xy}$ as well.

Vortices

$$H_{xy} = J\int dx\,dy\,(\vec{\nabla}\varphi)^2 = J\int dx\,dy\,(\partial_x\varphi)^2 + (\partial_y\varphi)^2$$

Spin correlation function at <u>low temperature $k_BT \ll J$</u>:

$$\Gamma(R) = \langle e^{i(\varphi(0)-\varphi(R))}\rangle = \int \prod_m d\varphi_m\, e^{i\varphi(0)-i\varphi(R)}\, e^{-\frac{J}{k_BT}\int d^2x\,(\nabla\varphi)^2} \Big/ Z$$

let us re-write the exponent: $Z = \int \prod d\varphi\, e^A$

$$A = \frac{J}{k_BT}\int d^2\vec{r}\left\{(\partial_x\varphi)^2 + (\partial_y\varphi)^2 - i\frac{k_BT}{J}\varphi(\vec{r})\left[\delta(\vec{r}) - \delta(\vec{r}-\vec{R})\right]\right\}$$

$$\varphi(\vec{r}) = \int \frac{d\vec{k}}{(2\pi)^2}\, e^{i\vec{k}\vec{r}}\,\varphi(\vec{k})$$

$$\int d^2\vec{r}\,(\vec{\nabla}\varphi)^2 = \int d^2\vec{r}\int\frac{d\vec{k}}{(2\pi)^2}\left(\vec{\nabla}\, e^{i\vec{k}\vec{r}}\,\varphi(\vec{k})\right)\int\frac{d\vec{k}'}{2\pi}\left(\vec{\nabla}\, e^{i\vec{k}'\vec{r}}\varphi(\vec{k}')\right)$$

$$= \int d^2\vec{r}\int\frac{d\vec{k}}{(2\pi)^2}\frac{d\vec{k}'}{(2\pi)^2}\,-\vec{k}\,\vec{k}'\,\varphi(k)\,\varphi(k')\,e^{i\vec{r}(\vec{k}+\vec{k}')} =$$

$g = \frac{k_BT}{2J}$

$$= \int\frac{d\vec{k}}{(2\pi)^2}\,\vec{k}^2\,\varphi(k)\,\varphi(-k)$$

$$A = -\frac{1}{2g}\int\frac{d\vec{k}}{(2\pi)^2}\left\{\vec{k}^2\,\varphi(k)\,\varphi(-k) - i2g\,\varphi(k)\left(1 - e^{i\vec{k}\vec{R}}\right)\right\}$$

Now we try to complete the square:

$$0 = \int d\vec{k}\left\{k^2\,\varphi(k)\,\varphi(-k) - ig\,f(k)\,\varphi(k) - ig\,f(-k)\,\varphi(-k)\right\}$$

where $f(k) = 1 - e^{i\vec{k}\vec{R}}$

$$0 = \int d\vec{k} \left\{ k^2 \left(\varphi(k) - \frac{ig}{k^2} f(-k)\right)\left(\varphi(-k) - \frac{ig}{k^2} f(k)\right) + k^2 \frac{g^2}{k^4} f(-k) f(k) \right\}$$

$$\Rightarrow \Gamma(R) = \int \Pi\, d\varphi_k \; \exp - \frac{1}{2g} \int \frac{d\vec{k}}{(2\pi)^2} k^2 \left(\varphi(k) - \frac{ig}{k^2} f(-k)\right)\left(\varphi(-k) - \frac{ig}{k^2} f(k)\right)$$

$$\cdot \exp - \frac{1}{2g} g^2 \int \frac{d\vec{k}}{(2\pi)^2} \frac{f(k) f(-k)}{k^2} \cdot \frac{1}{Z}$$

where $Z = \Gamma(R; f \equiv 0)$

$\Rightarrow$ integral over $\varphi_k$ factorizes and is canceled by the denominator $Z = \int \Pi\, d\varphi_k \exp - \frac{1}{2g} \int \frac{d\vec{k}}{(2\pi)^2} k^2 \varphi_k \varphi_k$

$$\Rightarrow \Gamma(R) = \exp - \frac{g}{2} \int \frac{d\vec{k}}{(2\pi)^2} \frac{(1 - e^{i\vec{k}\vec{R}})(1 - e^{-i\vec{k}\vec{R}})}{k^2} =$$

$$= \exp - g \int \frac{d\vec{k}}{(2\pi)^2} \frac{1 - \cos \vec{k}\cdot\vec{R}}{k^2}$$

Recall that
$$\lim_{a \to 0} \int_0^\infty \frac{dk}{k} \cos(ka\cos\beta) - \cos(kR\cos\beta) = \ln \frac{R}{a}$$

$$J = \int \frac{dk \cdot k}{(2\pi)^2} \underbrace{\int_0^{2\pi} d\beta \frac{1 - \cos(kR\cos\beta)}{k^2}}_{\frac{2\pi}{k^2}(1 - J_0(kR))} = \frac{1}{2\pi} \underbrace{\int \frac{dk}{k} \left(1 - J_0(kR)\right)}_{\approx \ln \frac{R}{a}}$$

<u>power law decay!</u>

$$\Gamma(R) \approx \exp - \frac{g}{2\pi} \ln \frac{R}{a} = \left(\frac{a}{R}\right)^{g/2\pi}$$

$$= \left(\frac{a}{R}\right)^{\frac{k_B T}{4\pi J}} \xrightarrow[R \to \infty]{} 0$$

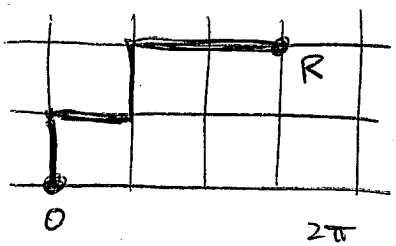
$R$ = distance between spins;

$a$ = distance between nearest sites (lattice spacing)

High temperature $k_B T \gg J$:

$$\Gamma(R) = \left\langle e^{i\varphi(0) - i\varphi(R)} \right\rangle = \frac{1}{Z} \int \prod_m d\varphi_m \, e^{i\varphi(0) - i\varphi(R)} \exp \frac{J}{2k_B T} \sum_{ng} \cos(\varphi_n - \varphi_{n+g})$$

Expand in powers of $\frac{J}{k_B T} \ll 1$.




Observe that $\int_0^{2\pi} d\varphi_m = 2\pi$, $\int_0^{2\pi} d\varphi_m \, e^{i\varphi_m} = 0$

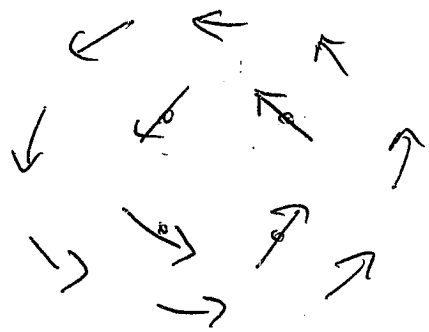
$\Rightarrow$ first contribution appears in order $\underline{\underline{R}}$

$$\Gamma(R) \sim \left(\frac{J}{2k_B T}\right)^{|R|} = \exp\left(-R \ln \frac{2k_B T}{J}\right).$$

<u>exponential decay!</u>

$\Rightarrow$ there must be transition between two regimes.
Kosterlitz and Thouless (also, Berezinskii in Russia) (1973)
this transition is topological, consists in
binding of <u>vortices</u>

<u>Vortex</u>
circulation

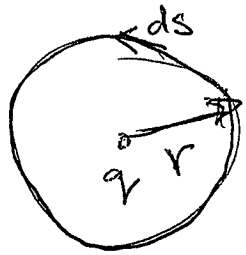


$$\oint d\varphi(r) = 2\pi q$$

for vortex "charge" $q$.

Implies that $\vec{\nabla}\varphi(r) = \frac{q}{r}\hat{\varphi}$

then $\oint ds \nabla\varphi(r) = r \cdot \frac{q}{r}\int_0^{2\pi} d\varphi = 2\pi q$




The energy of vortex

$$H = J\int d^2r \left(\vec{\nabla}\varphi\right)^2 = J\int d^2r \left(\frac{q}{r}\right)^2 = 2\pi J q^2 \int \frac{dr \cdot r}{r^2} =$$

$$= 2\pi J q^2 \ln\left(\frac{L}{a}\right)$$ where $L$ = size of the system.

But energy of <u>vortex pair</u> is finite (in $L \to \infty$ limit), depends on distance $R$ between vortices

$$H_{pair} \sim J \ln \frac{R}{a}$$

<u>Estimate</u> Entropy $S_{vortex} \sim k_B \ln\left(\frac{L}{a}\right)^2$

Free energy $F = E - TS = 2\pi J \ln\frac{L}{a} - 2k_B T \ln\left(\frac{L}{a}\right)^2$

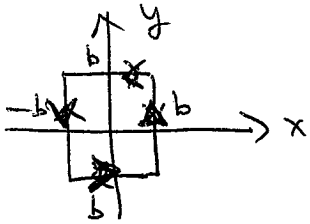
$$\Rightarrow T_c : \quad 2\pi J = 4 k_B T_c$$

$$\boxed{T_c = \frac{\pi J}{2 k_B}}$$

The key difference between $\cos(\varphi_r - \varphi_{r+g})$ and $(\vec{\nabla}\varphi(r))^2$ descriptions is that first form takes into account <u>compact</u> nature of the phase $\varphi_r$: changing $\varphi_r \to \varphi_r + 2\pi$ leaves cosine unchanged.
However, for $(\vec{\nabla}\varphi)^2$ such a change represents singular point (vortex).

Consider $\varphi_v(r) = q \tan^{-1}\left(\frac{y}{x}\right)$ and

calculate circulation around the origin:




$$\oint d\varphi_v = \int_{\mathcal{O}} ds \cdot \vec{\nabla}\varphi_v$$

$$\vec{\nabla}\varphi_v = (\partial_x \varphi_v)\hat{x} + (\partial_y \varphi_v)\hat{y} =$$

$$= -\frac{y}{r^2}\hat{x} + \frac{x}{r^2}\hat{y}$$

$$\oint ds \cdot \vec{\nabla}\varphi_v = \int_{+b}^{-b} dy\,(-\hat{y})\,\vec{\nabla}\varphi_v\Big|_{x=b} + \int_{b}^{-b} dx\,(-\hat{x})\,\vec{\nabla}\varphi_v\Big|_{y=+b}$$

$$+ \int_{+b}^{-b} dy\,\hat{y}\,\vec{\nabla}\varphi_v\Big|_{x=-b} + \int_{-b}^{b} dx\,\hat{x}\,\vec{\nabla}\varphi_v\Big|_{y=-b}$$

Clearly all 4 contributions are equal, evaluate one:

$$\int_{-b}^{+b} dy\left(-\frac{x}{r^2}\right)\Big|_{x=b} = -b\int_{-b}^{+b}\frac{dy}{b^2+y^2} = b\int_{-b}^{b} dy\left(\frac{1}{y+ib} - \frac{1}{y-ib}\right)\frac{+1}{2ib}$$

$$= \frac{+1}{2i}\ln\frac{y+ib}{y-ib}\Big|_{-b}^{b} = +\frac{1}{2i}\left(\ln\frac{b+ib}{b-ib} - \ln\frac{-b+ib}{-b-ib}\right) =$$

$$= +\frac{1}{i}\ln\frac{1+i}{1-i} = -i\ln\frac{e^{i\pi/4}}{e^{-i\pi/4}} = -i\ln e^{i\pi/2} = +\frac{\pi}{2}$$

$$\Rightarrow \oint d\varphi_v = 4\left(+\frac{\pi}{2}\right) = +2\pi$$

$\Rightarrow \varphi_v = \tan^{-1}\left(\frac{y}{x}\right)$ describes vortex of charge $(+1)$.

(convention here: positive circulation is counter clockwise)

Define $\overline{\varphi}$ : $\varepsilon_{\mu\nu}\partial_\nu \overline{\varphi} = \partial_\mu \varphi$

$$\partial_x \varphi = \partial_y \overline{\varphi} = \frac{-y}{r^2}$$

$$\partial_y \varphi = -\partial_x \overline{\varphi} = \frac{x}{r^2}$$

$$\Rightarrow (1)\ (\vec{\nabla}\varphi)^2 = (\partial_x \varphi)^2 + (\partial_y \varphi)^2 = (\partial_y \overline{\varphi})^2 + (\partial_x \overline{\varphi})^2 = (\vec{\nabla}\overline{\varphi})^2$$

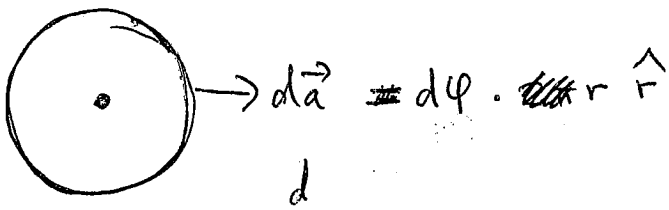
$$(2)\ \vec{\nabla}^2 \overline{\varphi} = \partial_x(\partial_x \overline{\varphi}) + \partial_y(\partial_y \overline{\varphi}) = \partial_x\left(\frac{-x}{r^2}\right) + \partial_y\left(\frac{-y}{r^2}\right)$$

$$= -\vec{\nabla}\cdot\frac{\vec{r}}{r^2} = -2\pi\,\delta^2(\vec{r})$$

Because by divergence Theorem: $\displaystyle\int_{\text{Volume}} (\vec{\nabla}\cdot\vec{A})\,d\tau = \oint_{\text{Surface}} \vec{A}\cdot d\vec{a}$

$$\Rightarrow \int (\nabla^2\overline{\varphi})\,d\tau = \oint -\frac{\vec{r}}{r^2}\cdot d\vec{a} = -\frac{\vec{r}\cdot\hat{r}}{r^2}\, r \int_0^{2\pi} d\varphi$$

$$= -\frac{r^2}{r^2}\cdot 2\pi = -2\pi$$




$$\Rightarrow \nabla^2\overline{\varphi} = -2\pi\,\delta(\vec{r}) \text{ indeed.}$$